Ruprecht-Karls-Universität Heidelberg
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Bachelor Thesis

Parallel ASP Planning

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Ich versichere, dass ich diese Bachelor-Arbeit selbstständig verfasst und nur die angegebenen Quellen und Hilfsmittel verwendet habe.

Heidelberg, dd.mm.yyyy

Zusammenfassung

Dies ist eine Zusammenfassung der Arbeit.

Abstract

This is the abstract.

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1 Einleitung

This chapter contains an overview of the topic as well as the goals and contributions of your work.

1.1 Motivation

1.2 Ziele der Arbeit

1.3 Aufbau der Arbeit

Here you describe the structure of the thesis. For example:

In Kapitel 2 werden grundlegende Methoden für diese Arbeit vorgestellt.

2 Basics

2.1 Definitions

Definition (multivalued planning task)**.** A STRIPS-like multivalued planning task according to (Helmert 2006) is given by a 4 tuple $\langle \mathcal{F}, s_0, s_*, \mathcal{O} \rangle$, where:

- $\mathcal{F}$ is a finite set of state variables, also called **fluents**. Each $x \in \mathcal{F}$ has a finite domain x^d and in every state, x takes exactly one value from its domain.
- s_0, the **initial state**, is a total function, s.t. $s_0(x) \in x^d$ for each $x \in \mathcal{F}$
- s_*, the **goal conditions**, is a partial function, s.t. $s_*(x) \in x^d$, for each $x \in dom(s_*)$
- $\mathcal{O}$ is a finite set of operators, also called **actions**. We define an action as $a = \langle a^c, a^e \rangle$, where a^c is the precondition of a and a^e is the postcondition of a. Both a^c and a^e are partial variable assignments over $\mathcal{F}$

Definition (successor state)**.** Let s be state and $a \in \mathcal{O}$ an action.
The **successor state** $o(a, s)$, obtained by applying action a in state s, is defined if the precondtion of a holds, i.e., $a^c(x) = s(x)$ for all $x \in dom(a^c)$. Then

$$s'(x) = o(a, s) = \begin{cases} a^e(x) & \text{if } x \in dom(a^e) \\ s(x) & \text{otherwise.} \end{cases}$$

Remark. Let $A_n = \langle a_1, \ldots, a_n \rangle$ be a sequence of actions with length n. We define the application of A_n to a state s recursevily by

$$o(A_n, s)) = o(a_n, o(A_{n-1}, s))$$

if $o(A_i, s)$ is defined for all $1 \leq i \leq n$

Definition (sequential plan)**.** A **sequential plan** is a sequence of actions $A_n = \langle a_1, \ldots, a_n \rangle$, s.t. $s' = o(A_n, s)$ is defined and $s'(x) = s_*(x)$ for all $x \in dom(s_*)$.

Bibliography

Helmert, M. (2006). “The Fast Downward Planning System”. In: *Journal of Artifical Intelligence Research* 26, pp. 191–246.